# Visual Question Answering on Chart Images

Multi-Scale CNN + Transformer + Cross-Attention Fusion

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> Teaching machines to understand charts and answer yes/no questions about colors, values, and relationships. I present a two-stream architecture combining a Multi-Scale CNN (or pretrained ResNet-18) with a from-scratch Transformer encoder, fused via stacked cross-attention, achieving **66.08% accuracy** on the FigureQA dataset — surpassing the original paper's baseline.

## 1. What is Visual Question Answering?

Given a chart image and a natural language question, predict **Yes** or **No**. The model must jointly understand visual content and language to reason about data encoded in charts.

**The core challenge:** Fine-grained color discrimination — is *"Dark Seafoam"* greater than *"Medium Seafoam"*? The model must distinguish between visually similar colors and map them to question tokens.

## 2. FigureQA Dataset

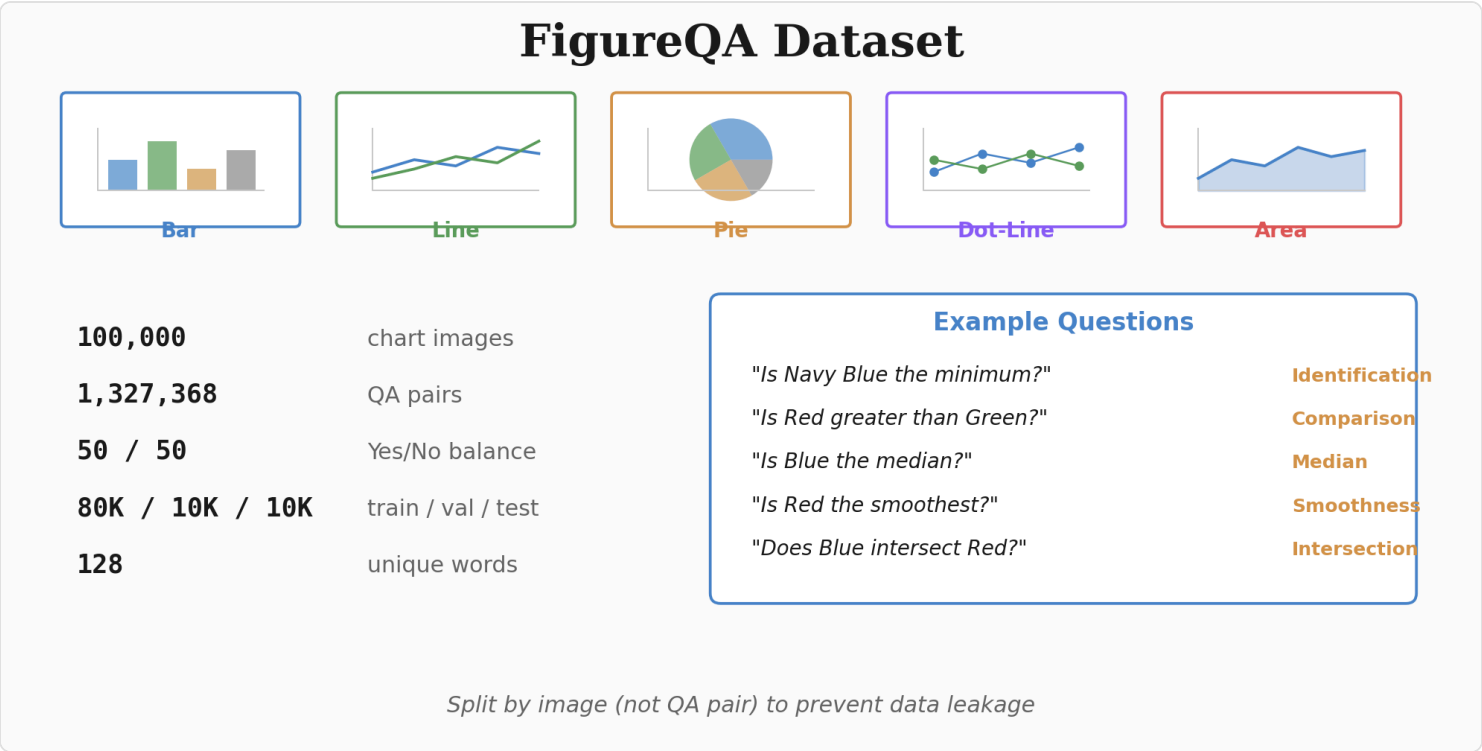



*Figure 0: FigureQA dataset — 100K chart images across 5 types, 1.3M yes/no question-answer pairs, split by image to prevent data leakage.*

| | |
|---|---|
| Chart Images | **100,000** |
| QA Pairs | **1,327,368** |
| Chart Types | **5** (bar, line, pie, dot-line, area) |
| Class Balance | **50/50** Yes/No |
| Image Splits | **80K / 10K / 10K** (train / val / test) |
| Vocabulary | **128** unique words (templated questions) |

## Question Types

FigureQA uses **15 templated question types** across two categories, requiring diverse visual reasoning beyond simple comparisons:

- **Identification:** *"Is X the minimum/maximum?"* — requires finding extremes across all chart elements
- **Comparison:** *"Is X greater/less than Y?"* — pairwise value comparison between two colors
- **Median:** *"Is X the median?"* — requires ranking all values to find the middle
- **Smoothness:** *"Is X the smoothest/roughest?"* — assessing curve regularity (line/area charts)
- **Intersection:** *"Does X intersect Y?"* — detecting line crossings (line charts)
- **Area under curve:** *"Does X have the largest/smallest area?"* — estimating cumulative values

**Why split by image?** Each chart generates ~13 QA pairs. Random splitting would leak visual features across train/val/test sets, inflating accuracy. I split by *image* to ensure the model never sees the same chart during evaluation.

# 3. Architecture Overview

A two-stream architecture that processes images and text in parallel, then fuses them through cross-attention for joint reasoning.

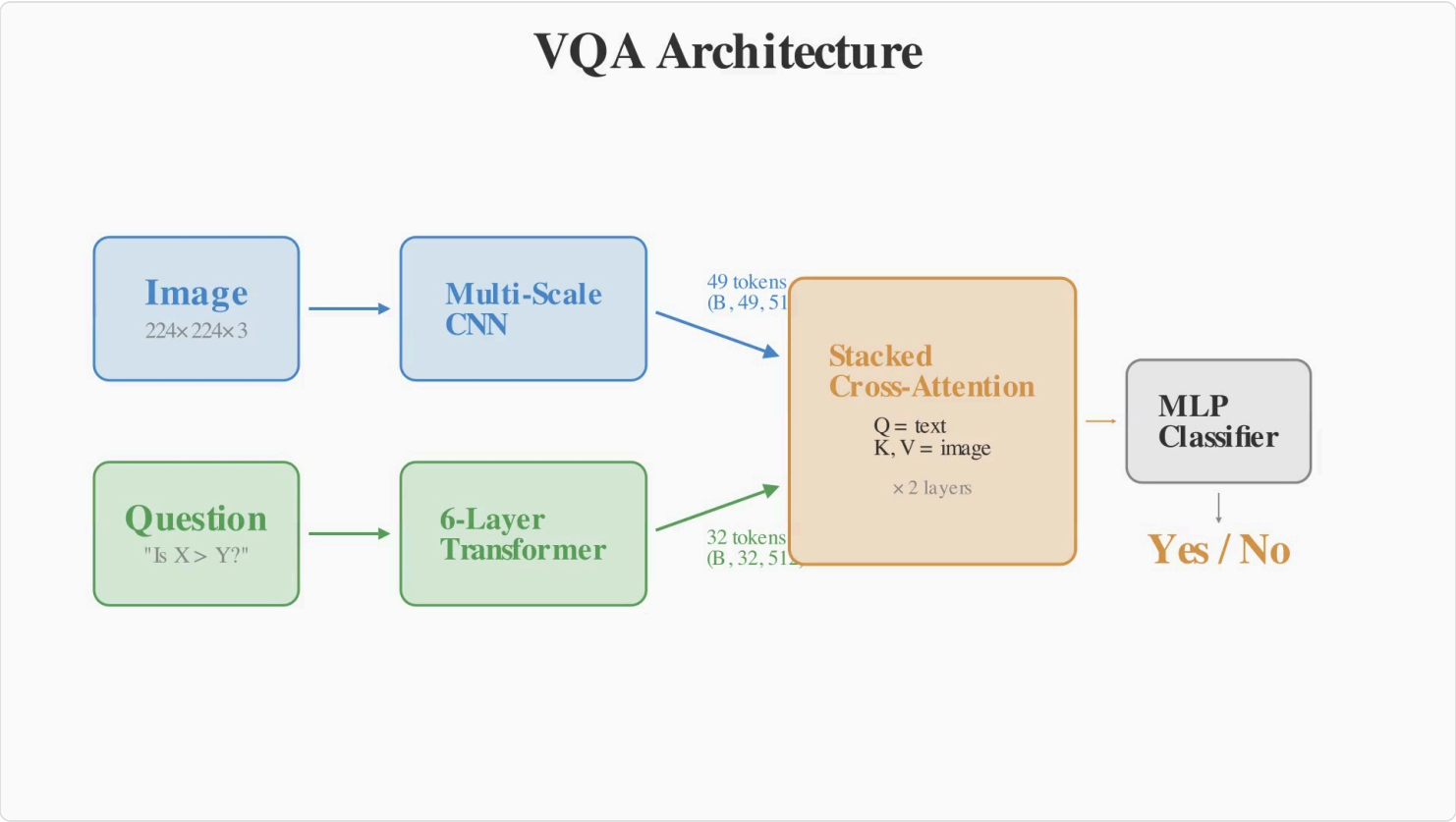



*Figure 1: VQA architecture data flow — image and question are encoded in parallel, fused via cross-attention, and classified by an MLP.*

21.9M parameters (from-scratch CNN) / 26.2M parameters (with ResNet-18)

# 4. Image Encoder: Multi-Scale CNN

A four-layer convolutional network that extracts features at multiple spatial scales, inspired by PReFIL's multi-scale approach.

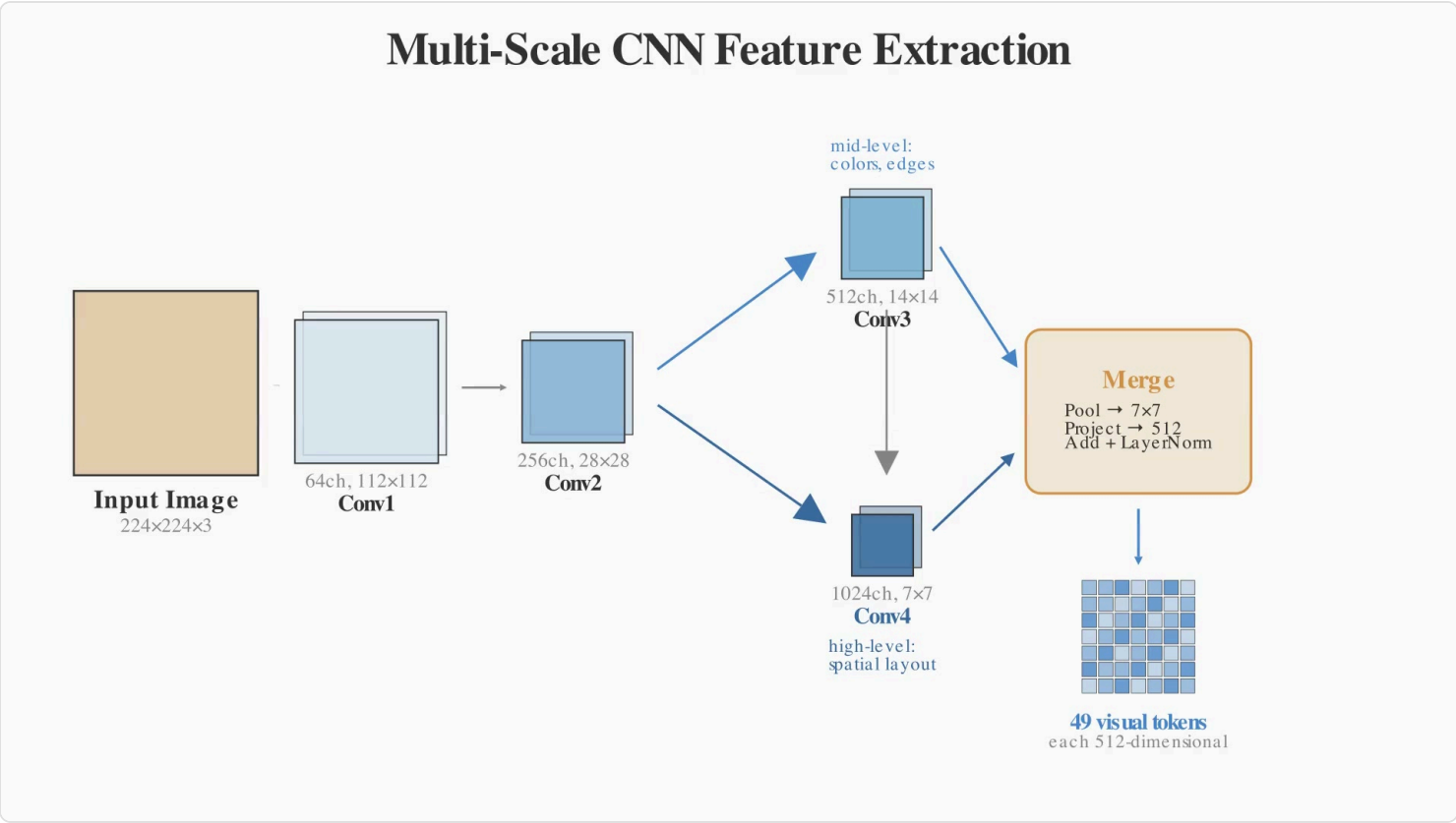



*Figure 2: Multi-scale CNN feature extraction — mid-level and high-level features are pooled and combined via element-wise addition.*

Mid-level features (Conv3) capture color information crucial for chart questions, while high-level features (Conv4) encode spatial relationships between chart elements. Both are pooled to a 7×7 grid, projected to 512 dimensions, and combined via element-wise addition with layer normalization.

## 5. Text Encoder: 6-Layer Transformer

A standard Transformer encoder built entirely from scratch — every attention head, projection, and layer norm hand-implemented.

Key design choices for the text encoder:

- **128-word vocabulary** — Templated questions mean a small vocabulary with no subword tokenization needed.
- **Sinusoidal positional encoding** — Deterministic, no extra trainable parameters.
- **Padding mask** — Variable-length questions handled with attention masking.
- **Fully from scratch** — Self-attention, projections, layer norms all hand-implemented.

# 6. Where Vision Meets Language

Cross-attention allows text tokens to query image features — each word learns which spatial regions of the chart are relevant to it.

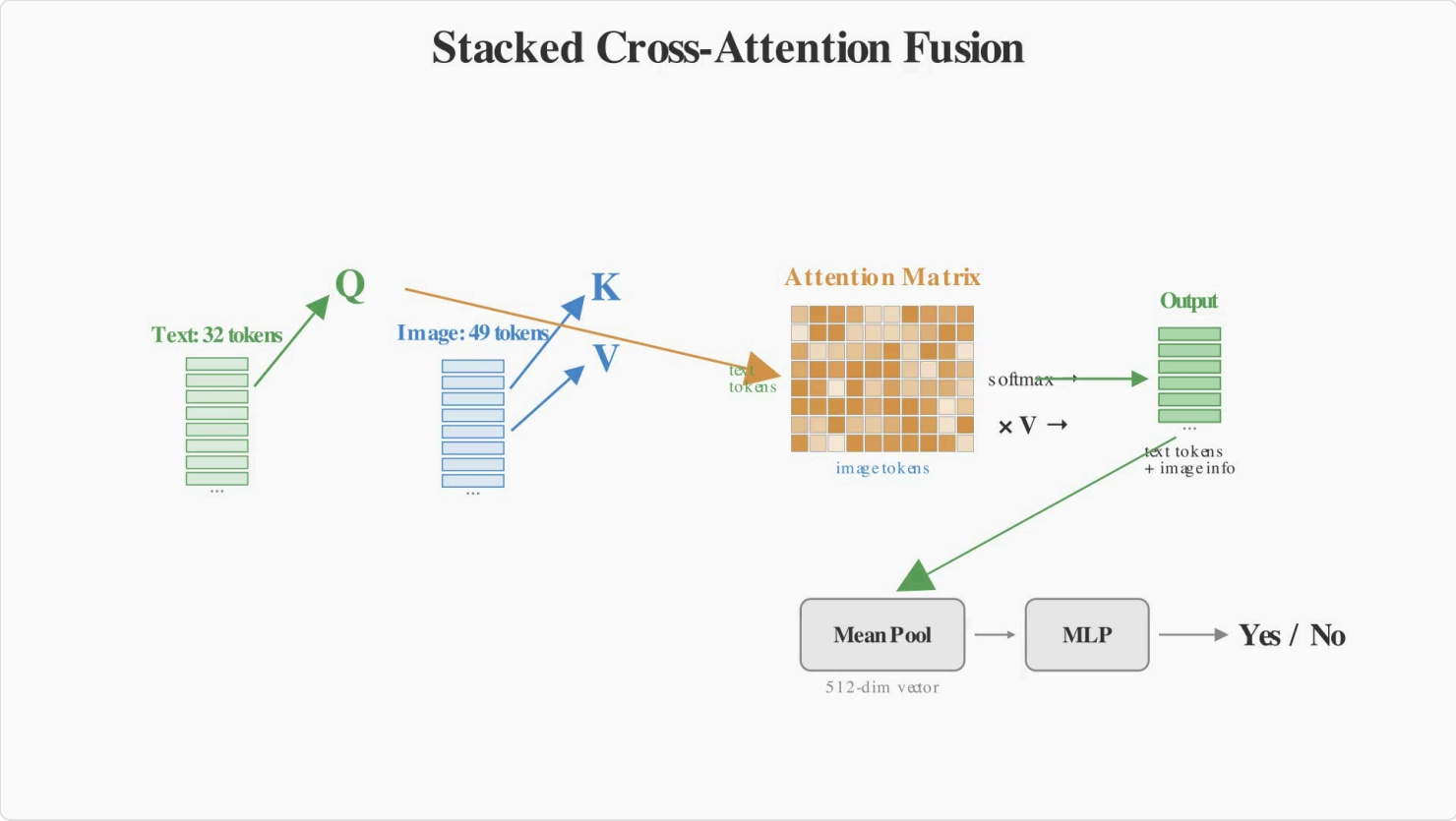



*Figure 3: Cross-attention fusion — text queries attend to image keys/values across two layers for multi-step reasoning.*

**Worked example: "Is navy blue greater than crimson?"** — Layer 1 locates both colors in the 7×7 spatial grid. Layer 2 compares their values. The fused representation is mean-pooled over the sequence to produce a single 512-dim vector, then classified via MLP(512→512→2) to predict Yes or No.

# 7. Training Setup

Two configurations: one with a from-scratch CNN and one with a pretrained ResNet-18 backbone. Same Transformer, same cross-attention, same MLP.

| | From-Scratch CNN | ResNet-18 |
|---|---|---|
| Image Encoder | Custom 4-layer CNN | Pretrained ResNet-18 |
| Parameters | 21.9M | 26.2M |
| Optimizer | AdamW (wd=0.01) | AdamW (wd=0.01) |
| Learning Rate | 3e-4 | 5e-4 |
| LR Schedule | Cosine + 1-epoch warmup | Cosine + 1-epoch warmup |
| Batch Size | 1024 (FP32) | 4096 (FP16) |
| GPU | A100 40GB | A100 80GB |
| Precision | FP32 | FP16 Mixed |
| Epochs | 10 | 10 |
| Gradient Clip | 1.0 | 1.0 |
| Label Smoothing | 0.1 | 0.1 |
| Dropout | 0.15 | 0.15 |

# 8. Regularization

Four complementary strategies to prevent overfitting on 1.06M training examples:

**Label Smoothing (0.1)** — Softens hard targets [1, 0] → [0.9, 0.1], preventing overconfident predictions and improving generalization.

**Dropout (0.15)** — Applied in self-attention, feed-forward network, and MLP classifier. Low enough to learn, high enough to regularize.

**Weight Decay (0.01)** — AdamW decoupled L2 regularization. Penalizes large weights without interfering with adaptive learning rates.

**Transfer Learning** *(ResNet only)* — ImageNet pretraining as implicit regularization. Pretrained features constrain the image encoder to a good region of weight space.

# 9. From-Scratch CNN Results

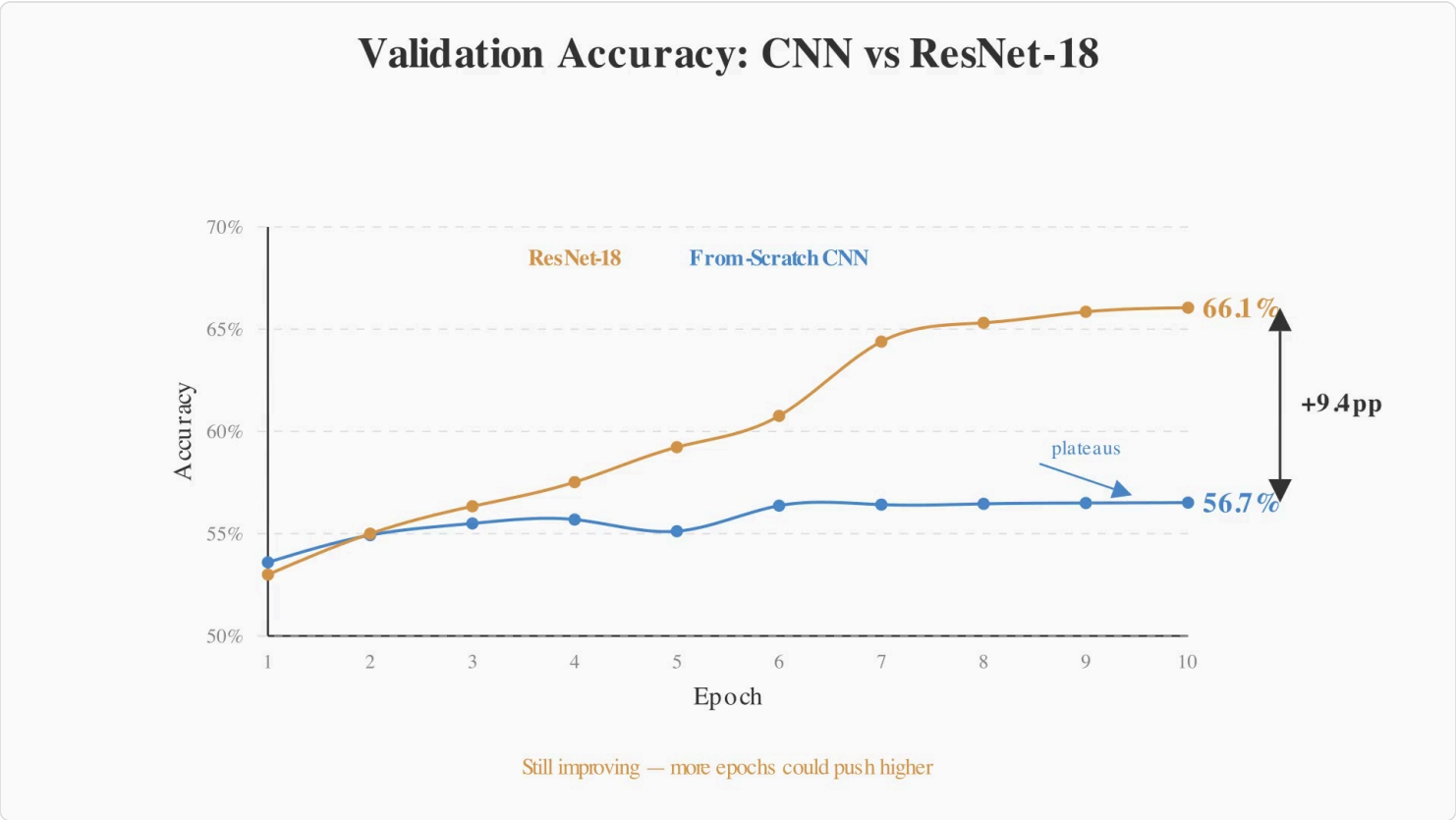



*Figure 4: Animated training comparison — the from-scratch CNN plateaus while ResNet-18 continues improving.*

The from-scratch CNN achieved **56.72% test accuracy**, with balanced per-class performance: **53.67%** on Yes questions and **59.76%** on No questions. The model plateaued around epochs 7–8, indicating no class bias but a fundamental ceiling.

The from-scratch CNN learns general spatial features but struggles with fine-grained color discrimination — the core challenge of chart VQA. With only 80K training images, four convolutional layers cannot learn the rich color representations needed to distinguish between "Dark Seafoam" and "Medium Seafoam."

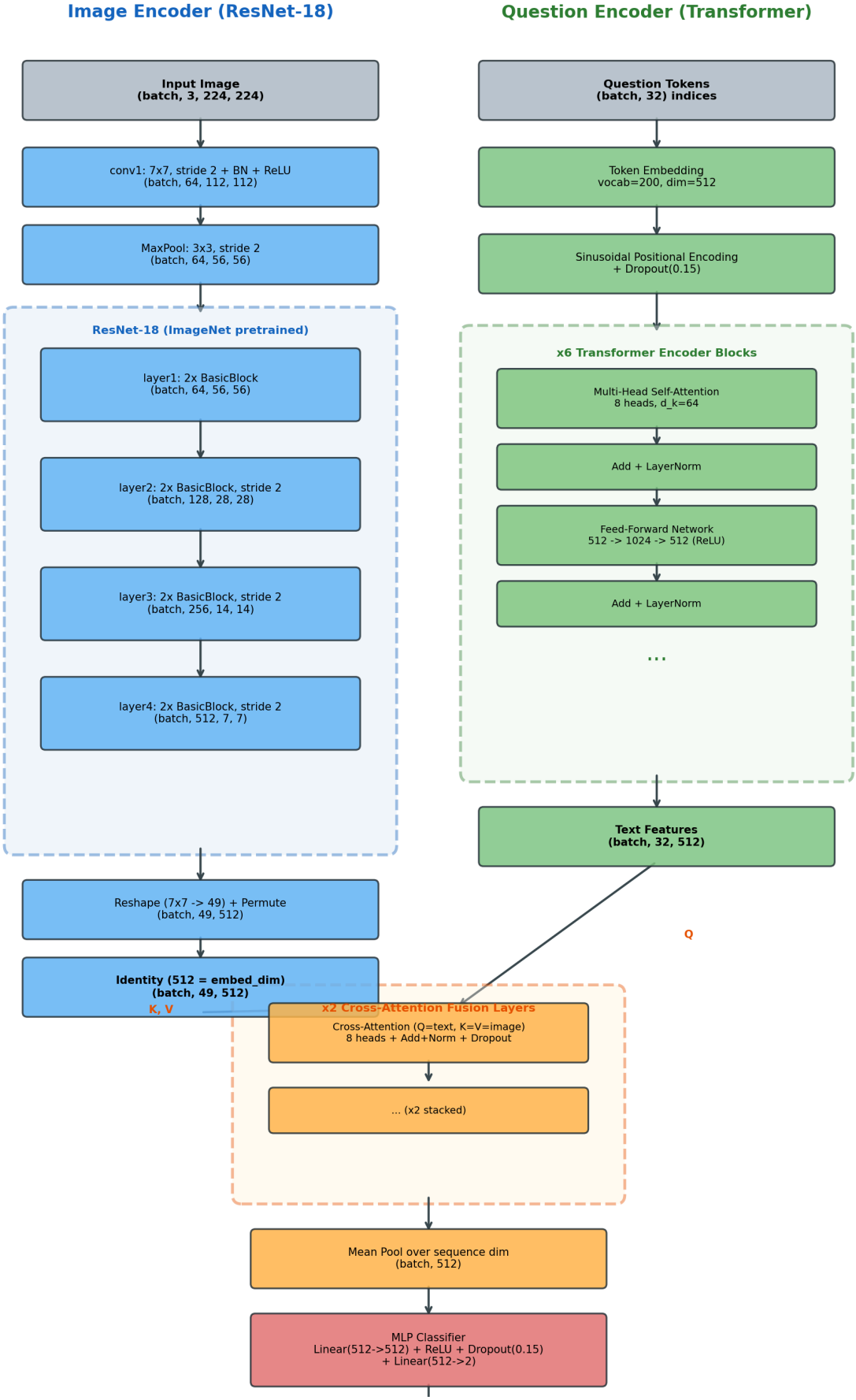
## 10. Transfer Learning: ImageNet → Chart Understanding

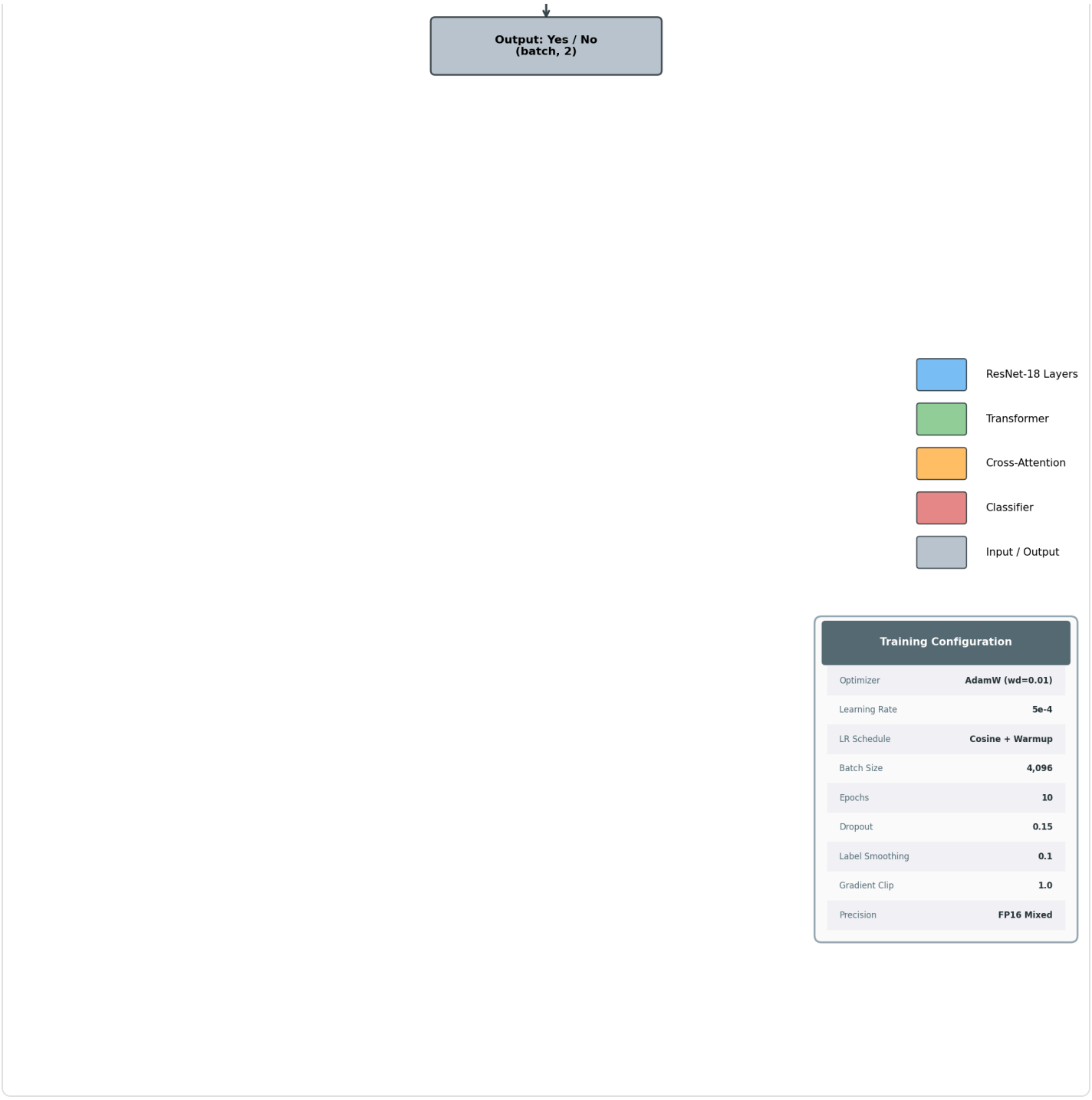
Why pretrained features transform chart VQA performance:

- **Rich visual priors** — ImageNet's 1.2M images of natural scenes teach the network to recognize colors, textures, and edges — exactly what chart VQA demands.
- **Data efficiency** — 80K chart images aren't enough to learn color representations from scratch. Pretrained weights provide a massive head start.
- **Low-level transfer** — Early CNN layers detect edges and color boundaries — these features transfer directly from photos to charts.

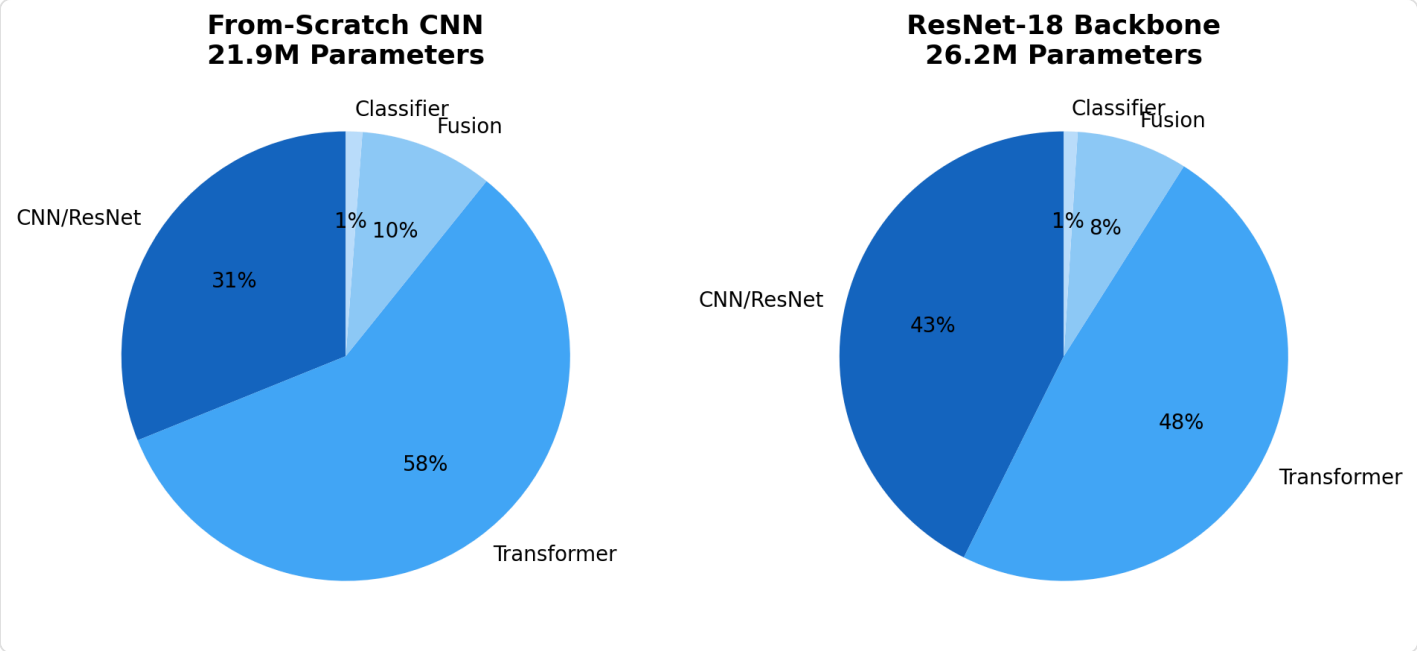
# ResNet-18 (Pretrained) + Transformer VQA

26.2M parameters | 66.08% test accuracy






*ResNet-18 backbone replaces the from-scratch CNN. Same Transformer + Cross-Attention + MLP.*



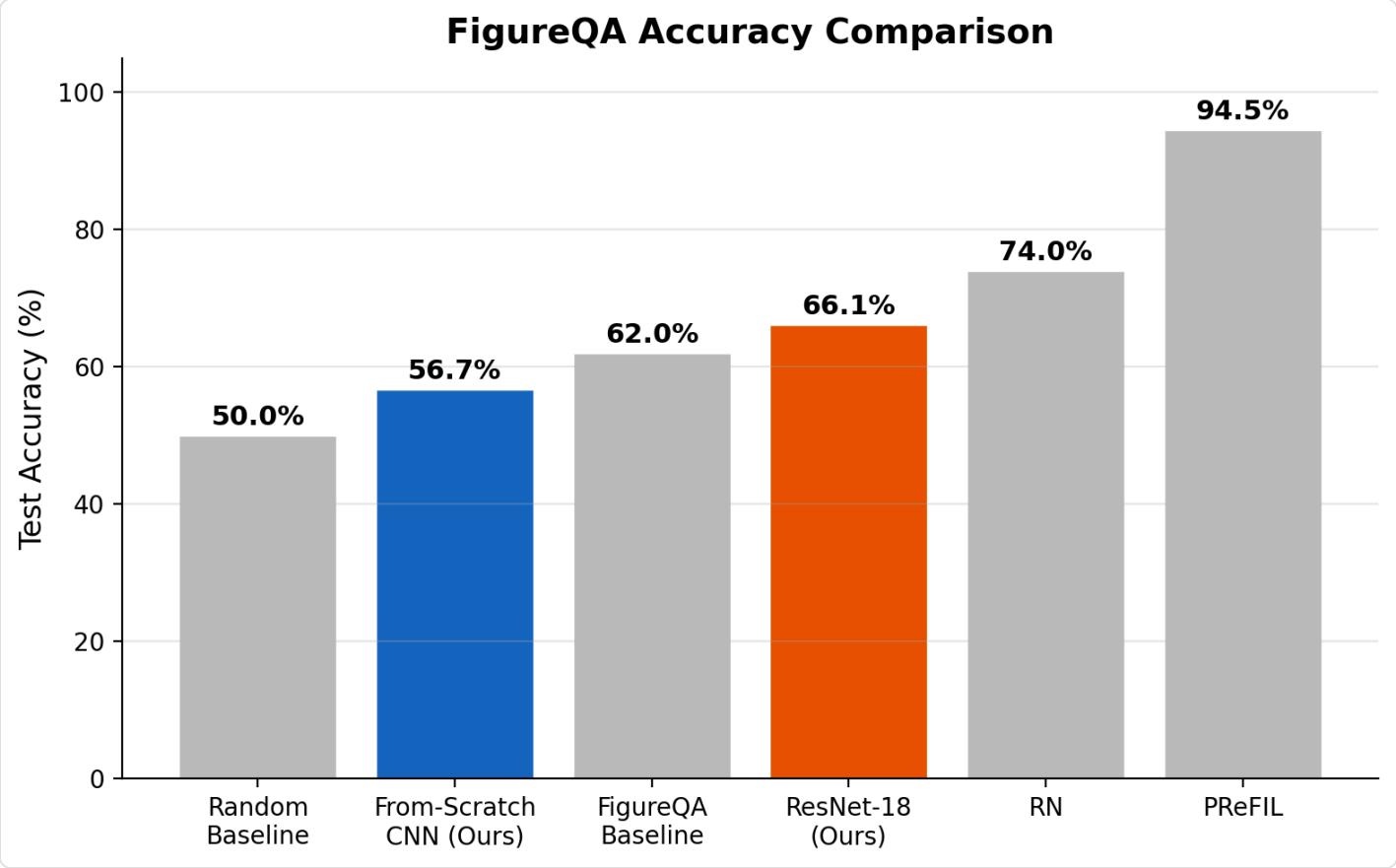

*Parameter distribution across model components — the text encoder dominates in both configurations.*

# 11. ResNet-18 Results

The ResNet-18 variant achieved **66.08% test accuracy** — a **+9.4 percentage point** improvement over the from-scratch CNN. The model was still improving at epoch 10 with no plateau, suggesting more epochs could push accuracy toward 70%+.

## Comparison with Benchmarks

| Model | Accuracy |
|---|---|
| Random baseline | 50.0% |
| **Ours (from-scratch CNN)** | **56.7%** |
| FigureQA paper baseline | ~62% |
| **Ours (ResNet-18)** | **66.1%** |
| Published SOTA (RN) | 72–76% |
| Published SOTA (PReFIL) | 93–96% |




*Accuracy comparison across baselines and my two model variants.*

## 12. Training Dynamics

**Warmup phase (epoch 1).** Learning rate ramps linearly from 0 to peak. The model reaches ~55% accuracy during warmup — necessary for training stability with large batch sizes and the Transformer's sensitivity to initial learning rates.

**CNN plateau (epochs 7–8).** The from-scratch CNN hits a 56.5% ceiling. This is a fundamental limitation: four convolutional layers trained on 80K images cannot learn the fine-grained color representations that chart VQA requires.

**ResNet keeps learning.** At 66.1% after 10 epochs with no plateau. The pretrained backbone enables continued improvement — more epochs could push accuracy toward 70%+.

**+15.4pp improvement on Yes questions** (53.7% → 69.1%), +3.3pp on No questions (59.8% → 63.1%). Transfer learning disproportionately helps affirmative reasoning — recognizing that a color *is* present requires finer visual discrimination than recognizing it is *not*.

# 13. Design Decisions

**Why 512 embedding dimension?** Matches ResNet-18's feature map width after adaptive pooling. Divides cleanly into 8 attention heads × 64 dimensions each — the standard configuration from "Attention Is All You Need."

**Why 2 cross-attention layers?** Layer 1 locates relevant image regions for each question token. Layer 2 performs multi-step reasoning — comparing values between located regions. One layer can find colors; two layers can compare them.

**Why multi-scale CNN?** Mid-level features (Conv3) capture color and edge information crucial for identifying chart elements. High-level features (Conv4) encode spatial layout and relationships between bars, lines, and segments.

**Why split data by image?** Each chart generates ~13 QA pairs. Random splitting would place questions about the same chart in both train and test sets, leaking visual features and inflating accuracy by up to 10%.

**Why mean-pool the fused features?** Simple aggregation over the sequence dimension. Fewer parameters than attention pooling or [CLS] token approaches, and empirically competitive for binary classification tasks.

**Why sinusoidal positional encoding?** Deterministic, requires no extra trainable parameters. With only 128 vocabulary words and 32 max sequence length, learned encodings offer no advantage over the analytical sinusoidal formula.

## 14. Conclusion

**Transfer learning works.** +9.4 percentage points from ImageNet pretraining. Pretrained color and texture features transfer directly to chart understanding, beating the FigureQA paper baseline.

**Cross-attention fuses modalities.** Two-layer cross-attention successfully bridges vision and language. Text tokens learn to attend to spatially relevant image regions for multi-step reasoning.

**Color is the bottleneck.** Fine-grained color discrimination remains the core challenge. The gap between 66% and SOTA 93%+ is primarily about distinguishing visually similar chart colors.

## Future Work

- **More epochs** — ResNet was still improving at epoch 10. Training for 20–30 epochs could push accuracy toward 70%+.
- **Larger backbones** — ResNet-50 or Vision Transformer (ViT) for richer visual representations.
- **Open-ended QA** — Extend beyond binary Yes/No to free-form answers about chart values and trends.
- **Attention visualization** — Overlay cross-attention weights on charts to show which regions the model attends to for each question.

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**Built by Ahmad Agah**

GitHub

Made with PyTorch, Manim, and too much coffee.